Register, Not Procedure: What a Conditional Linear Map Can and Cannot Install

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Abstract

Activation steering and parameter-efficient finetuning both edit a model's behavior, and both can be written as an *input-conditional affine map* of the residual stream, $h \mapsto h + (Wh + b)$. We hold this one map family fixed and sweep the *task*, asking a single question under one protocol: does the capability install? The boundary is sharp and reproducible. The map installs and transports a **register** — refusal tone, an answer format — reaching the harmful-refusal axis where every fixed-vector baseline stays at zero, and installing a commonsense donor's response format essentially completely (0.97 at one layer against a 0.004 base). Yet the identical machinery recovers $\approx$0 of a reasoning LoRA's gain on either multi-step **procedure** we test — GSM8K arithmetic and open-book multi-hop QA — as do MLP, on-policy DAgger, per-context, and task-loss (DAS) variants. The null is narrower than it first appears and we state it in the form the evidence supports: *a procedure does not transport through a fitted pointwise map at a single layer*. Concurrent work reaches finetuning-level reasoning with adapters trained on the task and applied at every block; our mechanism analysis says why that is the shape the problem requires: the procedure is a distributed, time-dense trajectory state (recovery needs $\geq 94\%$ of decode steps and $\sim 80\%$ of the residual's variance band), not a pointwise function of the activation. Running the full battery on a register task closes the loop from the other side: the register's full-state oracle has an earlier, sharper onset than either procedure's, its map transports the format but *destroys the answer selection the base already had*, and no vector estimated once — per-problem or pooled — installs anything at all. Every null is anchored to a positive control on the same instrument, so a null here means *not installable this way*, not *we did not try hard enough*. All results are on the Llama-2-7B family; we state this limitation, and the register/procedure criterion we propose, precisely enough to be falsified.

1 Introduction: "is X steerable" is not one question

The steering literature asks whether a behavior can be installed by editing activations, and tends to treat the answer as a property of the *method* — a better vector, a learned gate, a higher-rank edit. This paper argues the answer is dominated by a property of the *behavior*. We fix a single instrument — a closed-form, input-conditional affine map of the residual stream, the same object that parameter-efficient finetuning learns — and point it, unchanged, at behaviors of two kinds. A **register** is a disposition legible at each position: a refusal tone, a response format, an answer template. A **procedure** is a multi-step trajectory whose steps consume what earlier steps produced: chain-of-thought arithmetic, multi-hop composition.

Held to one protocol, the two kinds separate on every axis we measure, and the separation survives its own controls. The map installs registers (§4, §9) and never installs procedures (§6, §8); the full-state oracle that anchors every null recovers procedures only when it is distributed and temporally dense (§7), and recovers a register earlier, more sharply, and more completely (§9); and where the map *does* leak on a procedure, the leaked component turns out to be the task-agnostic register push, not the procedure (§8).

At a venue whose stated topics are measurement validity and falsifiability, the design commitment matters as much as the findings. Every null in this paper is anchored to a positive control on the same driver and scored on the same contrast set, so a zero is evidence about the behavior, not about our effort. Where our own controls destroyed a claim, the retraction is reported in line (§9), because the boundary we propose is only useful if it is stated in a form that can lose.

Contributions: (i) a two-sided, positive-controlled measurement of the register/procedure boundary on one instrument (§§4–9); (ii) a mechanism account of the procedure side — distributed, time-dense, variance-dense — that predicts which interventions can work and matches concurrent results that do (§7, §2); (iii) a register battery showing the boundary is visible in the *oracle* itself, not only in the map, and that register installation by weight-route edits costs selection the base already had (§9); (iv) a measured criterion table for "installability" with every cell sourced or marked unmeasured (§10).

2 Related work

Steering by a fixed direction. The dominant paradigm adds a single vector to the residual stream. CAA [Rimsky et al., 2024] takes the mean activation difference over contrastive pairs; ActAdd [Turner et al., 2023] obtains a direction from a pair of natural-language prompts; ITI [Li et al., 2023] shifts along learned directions at selected attention heads; RepE [Zou et al., 2023] reads and controls population-level directions for high-level concepts. For refusal specifically, Arditi et al. [2024] show a *one-dimensional* subspace mediates the behaviour across thirteen chat models. These methods share a commitment we make explicit and then test: the intervention is a **fixed offset**, identical for every input. §4 measures this family against a conditional map on the same prompts and grids, and §9 measures it on a second register with the vector estimated per-problem and pooled.

Conditional and matrix-valued steering. The fixed-offset assumption has been relaxed several ways, and the resulting family — an input-dependent affine edit of the residual stream — is occupied ground rather than something we introduce. CAST [Lee et al., 2025] gates a CAA vector with an explicit logistic classifier, making conditionality a bolted-on component. Conceptors [Postmus and Abreu, 2024] replace the vector with a *matrix*: the steered activation is $h' = \beta_c C h$, so the shift genuinely depends on the input, and $C = R(R + \alpha^{-2} I)^{-1}$ with $R = X^\top X/n$ is obtained in closed form. CLAS [Hsu et al., 2026] learns an input-dependent *coefficient*, $h \mapsto h + (c \cdot [h\ 1])\, d$, fit by gradient descent on next-token loss. ACE [Marshall et al., 2024] decomposes activations affinely, shows prior steering methods are subsets of terms of that decomposition, and controls refusal across ten models up to 70B. INNSteer [Nguyen and Le, 2026] goes further, learning invertible nonlinear transforms that induce input-dependent updates.

Two distinctions matter for what follows. CLAS is **rank-one conditional** — the direction d is fixed and only its magnitude adapts — where the map we study is full-rank, so its output direction varies with the input. Conceptors are matrix-valued and closed-form like ours, but differ in *target*: they take the second moment of a **single** activation set, yielding a shrinkage projection onto that set's ellipsoid with no regression target, whereas we use the **cross moment** $X^\top \delta$ of **paired** base and donor activations. That difference — supervision by a second model — is the only respect in which our instrument is unusual, and we treat it as an instrument rather than a contribution.

The register half of our result should therefore be read as consistent with this literature rather than as competing with it. ACE in particular reports the failure mode we also observe, that purely directional interventions produce incoherent output where affine ones do not, on far more models than we test. Our §4 exists as a **positive control**: it establishes that the apparatus we then point at procedures does install and transport a disposition.

Parameter-efficient finetuning as an activation edit. LoRA [Hu et al., 2022] learns a low-rank weight delta; LoReFT [Wu et al., 2024] learns an edit $h + (Wh + b - hR)R^\top$ applied to hidden representations of a frozen base. The latter is, formally, the same object as a conditional affine steering map, and the correspondence between the two spaces is not ours to claim: Adila et al. [2026] establish a first-order equivalence between activation-space interventions and weight-space updates, derive the conditions under which steering can replicate finetuning, and identify the post-block output as the expressive intervention site. Distributed Alignment Search [Geiger et al., 2024] supplies the complementary tool, finding task-relevant subspaces by gradient descent under a causal objective; it is one rung of our ladder in §6.

Why near-parity results are not counterexamples. Two concurrent papers report activation-space methods approaching finetuning, and both are compatible with our nulls for the same reason. Nguyen and Le [2026] reach within 3.64% of LoRA — on alignment traits, refusal and hallucination, and explicitly not on multi-step reasoning; that is our register side, independently confirmed, and their silence on procedures is the gap this paper fills. Adila et al. [2026] do evaluate GSM8K and AQuA and land within 0.2–0.9% of full-parameter tuning, which appears to contradict us directly. It does not, for three reasons that we make explicit because the surface similarity is otherwise misleading. First, **there is no donor**: every column of their comparison is trained on the task's own training split, so their claim is parity with supervised finetuning on the same data, not recovery of a capability that exists in another model. Ours is a transport question — whether a frozen base can be given a donor's capability — and no experiment in their paper addresses it. Second, our map receives **no task loss**; it is fit in closed form to reproduce a donor's activation delta. Third, their intervention spans **every block**, where the maps we falsify act pointwise at a single layer.

That third difference is the substantive one, and their results argue our case rather than against it. In their Table 1, ReFT — a single-site subspace edit, the closest published analogue of the map we study — is the weakest method reported, and degrades furthest exactly where the task is hardest for the model (GSM8K on Gemma-3-1B: 11.6 against 23.4 for supervised finetuning; ARC-Challenge 27.8 against 48.2; Winogrande on Qwen3-4B 64.9 against 83.0). Their remedy is to abandon the single site and intervene post-block at every layer. The ordering they measure — single-site subspace edit $\ll$ distributed post-block edit $\approx$ weight update — is precisely the ordering our §7 density measurements predict from the mechanism, obtained independently on four models we do not use. Their first-order analysis also isolates a term, $(\Delta W_d)m$, that activation steering cannot capture alone: a theoretical account of the wall we characterise empirically.

We therefore scope our claim accordingly, and state it in the form the evidence supports: a procedure does not **transport** through a fitted pointwise map at a layer. Installing one appears to require distributed, temporally dense intervention trained on the task — which is what Adila et al. [2026] supply, and what §7 independently shows to be necessary. All of Adila et al. [2026], Hsu et al. [2026] and Nguyen and Le [2026] are concurrent with this work.

3 Setup: one instrument, one protocol, positive controls throughout

The map. For a layer L we collect paired residuals (a, a^{donor}) from a frozen base and a donor (a task LoRA applied to the same base), and fit $W = \arg\min \sum \|Wa - \delta\|^2 + \lambda \|W\|_F^2$ in closed form (primal ridge over Gram accumulators), where $\delta = a^{\text{donor}} - a$. At inference the hook applies $h \mapsto h + \alpha W h$ at every position of layer L; no donor is present. We report held-out R^2 both raw and **centred** against the best-fixed-vector baseline $R^2_{\text{const}} = \|\sum \delta\|^2 / (n \sum \|\delta\|^2)$, because how constant δ is happens to be a property under study, and an uncentred R^2 credits a map for reproducing it.

The contrast-set protocol. Every task is scored on a cached *base-fails/donor-solves* contrast set (GSM8K: 113 problems of a 200-problem scan, base 0.000 / donor 0.565; MuSiQue: 317 of 500, $0.000/0.634$; ARC-Challenge commonsense: 338 of 500, $0.000/0.676$). Recovery is $(\text{acc} - \text{base})/(\text{donor} - \text{base})$. Because base scores 0.000 on its own failures *by construction*, a do-nothing intervention also scores 0.000; every null below is therefore paired with (i) a full-state **lockstep oracle** positive control on the same driver, and (ii) decoded generations, persisted in the run's artifact, distinguishing "no-op," "destroyed," and "base-like" — three readings a scalar zero cannot separate.

Table 1: Best harmful-refusal at benign over-refusal ≤ 0.10. Only the conditional map reaches the refusal axis. (Fixed-vector baselines reach no operating point on the frontier at any scale before coherence collapses.)

method	conditionality	harmful refusal	benign over-refusal
CAA mean-difference vector	none	0.00	—
Arditi single direction	none	0.00	—
CAA + logistic gate (CAST)	explicit classifier	0.00	—
ridge map Wa (α=0.12)	learned, for free	**0.62**	**0.02**

The oracle. The lockstep oracle overwrites the base's layer-L residual with the donor's, recomputed at every decode step on the base's own context. All-layers lockstep reproduces the donor's greedy output exactly on every task (the validation gate for every box and driver change); single-layer lockstep at L is the per-layer capability ceiling any pointwise-at-L method is bounded by. Layers 28/31 are excluded from L^* selection as degenerate (the hook overwrites the block *output*, so L31 is the all-layers control in disguise).

Reproducibility. All runs are seeded (42), all drivers carry CPU tests that run without network or GPU, and every number in this paper names its artifact in a provenance file (`numbers.md`) that maps claims to committed JSONs. Intervals are clustered bootstrap over *problems*, not tokens, since tokens within a chain are dependent.

4 (R) A register installs from the conditional map alone

Pointing the ridge map at refusal and scoring on the harm-refusal / over-refusal Pareto frontier (60+60 held-out prompts, refusal tone conditioned on coherence), only the *input-conditional* map reaches the harmful-refusal axis at all (Table 1). The fixed vectors go straight from no-effect to off-manifold gibberish (their per-layer ℓ_2 norm is $\sim$49); the map's per-input shift lands *on* the chat manifold and even repairs base coherence ($0.63 \rightarrow 0.98$) before it refuses. Conditionality is learned, not bolted on: Wa is large on harmful inputs and $\approx$0 on benign. The δ it regresses is near-perfectly decodable (held-out $R^2 \approx 0.9997$), so the open question was never "can δ be read off" but "does injecting it steer" — and for a register it does.

5 (R) Two routes to one register: LoRA and LoReFT diverge in the edit

A paper-faithful LoReFT (32 layers, rank 8, f7+l7) and a matched LoRA trained on the *byte-identical* supervised signal both recover commonsense QA from a 0.00 base (BoolQ/PIQA/ARC-C: LoReFT $0.667/0.798/0.602$, LoRA $0.680/0.790/0.655$). But the *edits* are not the same change: $\mathrm{CKA}(\delta)$ collapses 0.96 (L4) $\rightarrow$ 0.13 (L28), per-token cosine stays 0.11–0.39, and top-8 subspace overlap stays 0.16–0.25. They share only an early-layer, rotation-equivalent skeleton — the register/format nudge — while the task content is written through genuinely different deep-layer edits. There is no canonical "commonsense direction." §9 returns to this task with the full battery and shows the shared skeleton *is* the installable part.

6 (P) A procedure does not install through any pointwise map we tried

The same machinery pointed at GSM8K recovers $\approx$0 of a MetaMath-LoRA's budget [Yu et al., 2024], robustly across the obvious salvages (Table 2). The map installs reasoning-*shaped* tokens with no arithmetic underneath.

Each rung shares two properties with the others and with nothing else: the estimator is fit to reproduce a donor's activation delta rather than trained on the task, and it acts pointwise at a single layer. The null is therefore a null about **transport through a pointwise map**, and it should not be read as a claim that reasoning is beyond activation-space intervention in general — a reading that concurrent results refute. Adila et al. [2026] reach within 0.2–0.9% of full-parameter finetuning on GSM8K with an adapter trained on the task's own training split and applied post-block at every

Table 2: The procedure-null ladder on GSM8K. Every rung is fit to reproduce the donor's δ (or, for DAS, trained under task loss at one site) and acts pointwise at a single layer. Bracketed values are 95% bounds on the recovery scale: exact binomial for the zero rungs (treating base and donor accuracies as known constants, so they cover sampling error in the steered run only), clustered bootstrap over problems for the measured rungs.

salvage attempt	recovery @95%	rules out
global primal-ridge map (n=30; per-layer at L20/L24: §8)	0.00 $[0, 0.22]$	— (baseline)
short-output single-operation arithmetic (n=30)	0.00 $[0, 0.16]$	trajectory *length*
per-context local refit (problem-specific map, n=30)	0.00 $[0, 0.22]$	map *generality*
on-policy DAgger refit (≈97% on-policy by round 2, n=30) [Ross et al., 2011]	0.00 $[0, 0.23]$	*distribution shift*
MLP head (cos 0.815, R^2 0.675 at L24, n=100)	0.00 $[0, 0.04]$	*linearity* of the estimator
DAS task-loss subspace (CE→0.038, $r \in \{8, \ldots, 512\}$, n=20) [Geiger et al., 2024]	0.00 $[0, 0.17]$	subspace *search*

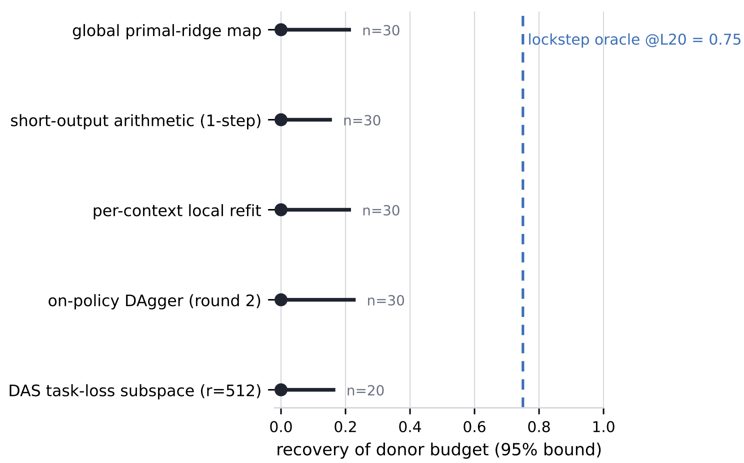

Figure 1: The procedure-null ladder against its positive control. Dots are recovery point estimates; whiskers are exact-binomial 95% bounds (steered run only). Every pointwise estimator sits at zero with a bounded ceiling; the lockstep oracle on the same contrast set recovers 0.75.

layer. The DAS rung is the one that most nearly bridges the two settings, since it is trained under a task objective; what still separates it is placement and capacity — rank 512 at one layer against an adapter spanning the whole stack — and placement is precisely what their analysis identifies as decisive. We take their result as evidence for the mechanism in §7 rather than against the null here: if the recoverable state is distributed across layers and dense in time, an intervention that succeeds must be distributed and dense, and theirs is. Consistent with this, the weakest method in their own comparison is ReFT — a single-site subspace edit, the closest published analogue of the map we falsify — which on the hardest settings falls far below supervised finetuning (GSM8K, Gemma-3-1B: 11.6 against 23.4).

One rung of this ladder moved when we measured it more carefully, and we report it as found: the linear rung is not exactly zero late in the stack. Per-layer steering at the two layers our original grid missed reads 0.031 $[0.01, 0.08]$ at L20 and 0.123 $[0.07, 0.19]$ at L24 (n=200); a re-fit over all positions and an α probe revise L20 upward to $\sim$0.09 and confirm L24 at 0.12 (both lower bounds — the curves still rise at the last α probed). §8 shows what this late-stack leak is made of; it is not the procedure.

7 Mechanism: the procedure is a distributed, time-dense trajectory state

A lockstep oracle that overwrites the base's L20 residual with the LoRA's every step *does* recover GSM8K (0.75), which lets us dissect what the map cannot supply.

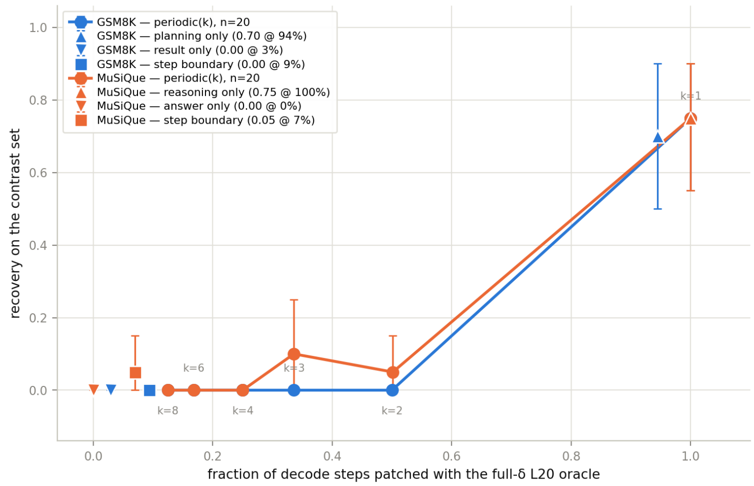

Figure 2: Temporal density of recovery. Periodic gates on the lockstep oracle: patching every step recovers the budget; every-other-step collapses it; sparser gates are zero. Both procedures show the same knee; the structural complement (patch everything except the answer span) equals the full oracle.

(i) Distributed, not transported. Identity-ablating just the top two downstream layers (30–31) craters not only the patched base ($0.75 \to 0.10$) but the natively-capable LoRA itself ($1.00 \to 0.10$) — L20 holds an *intermediate*, not a finished answer.

(ii) Compute is not the deficit. Teacher-forced on the correct chain, the base predicts computed-result tokens at 96.8%; its failure is *trajectory control*, not per-step arithmetic.

(iii) Dense in time and in variance. Recovery requires patching $\geq 94\%$ of decode steps (every $\leq 50\%$ gate $\to$ 0.00; Figure 2) and $\sim 80\%$ of δ's variance band (top-64 $=$ 55% of energy but 0% recovery; the cliff is top-256$\to$512). A donor-free steerer would therefore have to reproduce the LoRA's near-exact 4096-d state at almost every step — i.e., *be the donor*. The procedure is not a low-rank, pointwise, or time-sparse function of the activation.

8 (G) A second procedure: the scaffold generalizes, the per-step work does not

One procedure cannot separate "procedures do not install" from "arithmetic does not install," so we re-ran the apparatus end-to-end on MuSiQue open-book multi-hop QA [Trivedi et al., 2022] (matched LoRA $0.000 \to 0.634$; 317 base-fail/donor-solve contrast problems).

The two structural axes replicate exactly. *Oracle:* the lockstep oracle at the same layer recovers $+0.76$ (GSM8K 0.75), all-layer control $+1.00$. *Temporal density:* patching every other step collapses recovery to 0.05, every sparser periodic gate is ≈ 0, and the structural complement — patch everything except the answer span — equals the full oracle, the same signature GSM8K gives. *Plan vs. execute:* teacher-forcing the base on the gold chain and lensing the gold token by its role, the base agrees with **execution** tokens more than **planning** tokens ($+0.055$ [+0.040, +0.069], bootstrapped over problems), so the trajectory-control reading survives the task change — and supplying the trajectory makes the *later*, nominally-composed hops *easier* than the first ($+0.130$), which is what a trajectory deficit predicts and a per-step composition deficit does not. What does *not* carry over is the shape: where GSM8K's computed results sit $+0.133$ [+0.096, +0.173] above the chain average and crystallize with depth (lens rank $18 \to 7 \to 0$ over L20–24), multi-hop execution merely *matches* the chain average ($+0.001$, interval spanning 0) and no role crystallizes at all — its hop answers are copies from the in-context passages (the final restatement scores 0.933 at lens rank 0 in every layer). The trajectory-control deficit is general; the per-step work the base retains is task-specific, and only on arithmetic is it computation rather than retrieval.

Table 3: Single-layer lockstep oracle recovery by task (n=100 contrast problems each, same driver and protocol). Layers 28/31 are degenerate (block-output overwrite) and excluded from L^* selection. GSM8K's L20 value at n=100 (0.720) is consistent with the two prior n=20 measurements (0.75).

layer	commonsense (register)	MuSiQue	GSM8K
12	0.050	0.020	0.000
16	**0.830**	0.020	0.310
20	**0.990**	0.760	0.720
24	0.990	0.780	0.760

The ladder axis, matched at last — and what the leak is. The one place the two procedures appeared to differ was the linear rung: multi-hop's ridge map leaked 0.26 at L20 and 0.45 $[0.35, 0.56]$ at L24 where GSM8K's was believed to read ≈ 0. Both halves of that comparison were mismeasured, in opposite directions, and matching the measurement collapses most of the gap. GSM8K, probed at the two layers the original grid missed, also leaks late (0.09–0.12 at L20/L24; §6). And multi-hop's $0.26/0.45$ came from a map fit on chain positions only but applied everywhere — the fit-window mismatch of §9 in milder form. Re-fit over all positions and run under the identical protocol (n=200, same references), multi-hop reads 0.07 at L20 and 0.26 at L24. The matched comparison is therefore: **L20** 0.09 **vs.** 0.07 **— no task difference at all — and L24** 0.12 **vs.** 0.26, a factor of ~ 2 where the unmatched numbers suggested $9\times$ and $3.75\times$. All four cells are lower bounds (recovery still rises at the last α probed). The late-stack leak is real, small, and shared; its size differs across procedures by at most a factor of two. Two further measurements say what the leak is made of. **It is task-agnostic:** maps fit on *multi-hop* δs and applied to GSM8K deliver 75–100% of the native leak (L24 0.09 $[0.05, 0.16]$ vs. native 0.12; L28 0.13, equal to native exactly), so the task-fitted component transports ≤ 0.03 — the leak is a late-stack *register push*, not transported procedure. **And better fit means less transport:** at the live layer the MLP head fits δ better than ridge ($\cos 0.815$ vs. 0.656, R^2 0.675 vs. 0.354) and transports *nothing* (0.00 $[0, 0.04]$ vs. ridge 0.10 $[0.05, 0.18]$, disjoint intervals) — the transportable component is the *unconditional* one, which the better estimator correctly attributes to the conditional structure it cannot transport.

9 (R) The register battery: the other half of the boundary, measured

The claim so far is two-sided but the battery was one-sided: both procedures have an oracle sweep, a temporal knee, and a null ladder; the register side rested on refusal alone. We therefore pointed the *unmodified* procedure drivers at a register donor: a commonsense LoRA (r32, same recipe as §5) whose target is the $\sim$7-token template "`the correct answer is answerX`" on ARC-Challenge, with a 338-problem contrast set (base 0.000 / donor 0.676). ARC-Challenge is 4-way, so a garbled intervention cannot pass by coin-flip: the measured floors are 0.25 (chance) and 0.288 (majority class).

The oracle itself separates register from procedure. Under the same driver and the same earliest-plateau rule, all three tasks select L^*=20 — but the curves differ in exactly the way the two-kinds view predicts (Table 3, Figure 3). The register's onset is both earlier and nearly complete on arrival: 0.830 at L16, against GSM8K's partial 0.310 and MuSiQue's 0.020 at the same layer. Its plateau is essentially total (0.990 against $0.720/0.760$). The two procedures share a plateau to within noise (0.72–0.78 from L20) while differing in ramp shape — GSM8K climbs gradually through L16 where MuSiQue steps at L20 — so the register/procedure contrast is not a two-task idiosyncrasy: both procedures sit far below the register at L16 and far below its ceiling at the plateau. The onset comparison is within-task-across-layer and thus immune to generation-length effects; the ceiling comparison is not (the register's target is $\sim$7 tokens against 256–512-token chains), and we flag it accordingly.

The fitted map installs the register — completely, and at one layer. Fit on all positions (the fit window must match the application window; fitting on generated positions only and applying everywhere destroys the model — a failure mode we hit, diagnosed from decoded generations, and now test for), the ridge map's *format compliance* at α=0.75 on the 500-problem scan is: L16 0.140, **L20** 0.972, L24 0.998 (base 0.004, donor 1.000). Set beside the same instrument on the procedures,

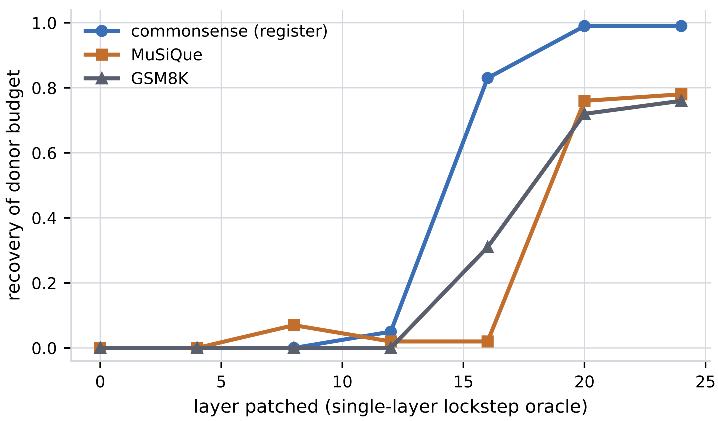

Figure 3: Single-layer lockstep oracle recovery by layer, all three tasks on one instrument. The register's onset is early and nearly total; both procedures plateau together at $\sim$0.72–0.78 from L20. Degenerate layers 28/31 excluded.

under matched all-positions fits — GSM8K $0.09/0.12$, MuSiQue $0.07/0.26$ at L20/L24 (§8) — this is the boundary on one axis: *a register transports through a fitted pointwise map at a single layer, essentially completely; a procedure leaks at most a quarter of its budget, late in the stack, and §8 shows even that fraction is a task-agnostic register push.*

It installs the register and destroys the selection the base already had. Under the map, accuracy equals the base rate of a constant policy — 0.200 with gold-`answer1` at $8/40$, replicated at 0.267 with $16/60$ — and there is no α that installs and selects: α=1.24, which magnitude-matches the donor's shift, collapses format itself to 0.083. Meanwhile the contrast set turns out to be $\sim$two-thirds *format*: base scores 0.630 with a 4-shot prompt, and a scrambled-label control also scores exactly 0.630, so the exemplars supply format and nothing else. Three interventions, one table: the CAA-style fixed vector installs neither format nor selection (0.000); the fitted map installs format and *destroys* selection (0.200, a constant policy); the 4-shot prompt installs the same format and leaves base's own selection intact (0.630). The weight-route register install costs something the prompt-route install does not. Consequently the oracle's 0.990 must not be described as recovering a capability base lacks — it is substantially format installation, and we say so.

No vector estimated once installs anything. A per-problem trajectory-mean vector through an additive hook: 0.000. The same vector through the oracle's own delivery path: 0.000. A pooled vector over 100 disjoint problems, at $\alpha \in \{0.5, 1, 1.5, 2\}$: 0.000. On the *temporal* axis, then, the register behaves like the procedures — a deployable-in-principle fixed vector installs nothing — and the boundary lives in the oracle's onset/ceiling and in the fitted map, not in "the register is one direction." It is not: the centred held-out R^2 of the register's map is 0.921 (constant baseline 0.106), the required shift rotates substantially within a 7-token generation, and a matched-norm *coherent* random direction — injected through the same plumbing that installs the register when given the mean-δ direction — reads 0.000 with base-like generations. A per-step statistic that collapses the shift across *positions* but keeps a live donor in the loop retains 0.860 of the oracle; collapsing across *time* is what kills it. Direction is everything; magnitude alone is nothing.

10 A measured criterion for installability

Table 4 assembles the coordinates this paper measured. We propose them as a falsifiable criterion: a behavior installs through a pointwise conditional map iff its required shift is (i) transportable at a single layer (map arm), (ii) recovered by the oracle with early, sharp onset, and (iii) not time-dense at the trajectory level. Cells we did not measure are marked — a criterion is only as good as its unmeasured cells are visible.

Two readings the table supports, and one it does not. It supports: registers sit in the top-left corner (high map transport, early sharp oracle onset) and procedures in the bottom-right (map transport at

Table 4: The criterion table. “n.m.” = not measured (stated, not estimated). Ridge cells for the procedures are lower bounds (§6).

	refusal	commonsense register	GSM8K	MuSiQue
map transport (single layer)	0.62 (Pareto)	0.972 (format)	0.09–0.12	0.07–0.26
oracle onset / plateau	n.m. / n.m.	L16 0.830 / 0.990	L16 0.310 / 0.720	L16 0.020 / 0.760
fixed vector (time-collapsed)	0.00	0.000	0.00	n.m.
centred held-out R^2 of δ	≈1.0	0.921	0.653	0.822
temporal density (periodic-2 gate)	n.m.	n.m. (7-token target)	→0.00	0.05
δ-rank / variance band	n.m.	n.m. (cut)	80% band needed	n.m.

leak level only, late onset, time- and variance-dense); and the coordinate that does *not* separate the two sides is the constancy of δ (R^2_{const} 0.096 vs. 0.106 at L20) — the discriminator is the *centred* R^2, i.e. how predictable the *conditional* part of the shift is. It does not support: any claim about behaviors outside this table. Sycophancy, sentiment, induction — the held-out behaviors that would turn this from a summary into a criterion — are future work, and the table's own empty cells show what measuring one costs.

11 (X) Cross-model: the disposition carries; better aim transports worse

Transplanting a trained LoReFT edit from a donor (Llama-2-7B base) to a recipient (Llama-2-7B-chat) through a per-layer orthogonal-Procrustes bridge transfers the *disposition* off-policy (BoolQ boolean-commitment $32 \to 53/64$, lenient $0.33 \to 0.58$) but not the trained output template or answer correctness (exact-match ≈ 0). On-policy refit is *worse*; and a ridge-fit **affine** bridge with 12–45× better held-out geometry transports the edit *worse still* ($8/64$), because the LoReFT delta lives off the data manifold ($\|\delta\| > \|h\|$) and regression shrinks weakly-represented directions — precisely the ones that must be preserved [Oozeer et al., 2025]. Strength beats aim: full-norm deltas through a crude isometry move behavior; well-aimed deltas at half norm do nothing. The geometric “wall” was real but behaviorally irrelevant — consistent with §8's finding that the better-fitting estimator transports less.

12 Synthesis and limitations

One axis organizes the results. A **register** — refusal tone, an answer format — is an on-manifold, input-conditional but *pointwise-transportable* function of the activation: a single-layer closed-form map installs it (0.62 Pareto; 0.972 format), and even a crude cross-model rotation carries its coarse disposition. A **procedure** is a distributed, time-dense, variance-dense *trajectory state*: invisible to every pointwise estimator we fit (the better the fit, the less transports), recoverable only by supplying the donor's near-full state at almost every step. Finetuning and steering are the same map family and inherit the same boundary: they relocate *where a disposition points*, not *how a computation unfolds*. And the boundary is visible from both sides of the same instrument: the register's oracle onset is early and sharp where the procedures' is late; the register's map installs the surface completely — at the price of the selection the base already had — where the procedures' map carries only a task-agnostic register push.

Limitations. (1) Everything here is Llama-2-7B; the apparatus requires only a Llama layer layout and a trained donor, so cross-model replication is mechanical, but it has not been done. (2) $n{=}2$ procedures and $n{=}2$ registers; the criterion table names what a third of each costs. (3) The register's oracle *ceiling* comparison carries a generation-length confound (7-token vs. 256–512-token targets); the onset claim does not, and we lead with onset. (4) The register-side geometry (map-vs-donor cosine 0.788; few-shot route $\sim 70°$ away) awaits a length-matched control before “format direction” language is licensed, and we do not use it. (5) Two criterion cells are lower bounds (α peaks not located) and several are unmeasured; they are marked in the table rather than extrapolated.

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